Newton's Laws

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Newton's Second Law ($F = ma$)**

Particles of mass 11 kg and 16 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 127.76. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 1.81$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 11(9.8 + 1.81) = 127.76$ N

2 **[Foundational]** • **Target: Connected Particles over a Pulley**

Particles of mass 9 kg and 22 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 125.19. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 4.11$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 9(9.8 + 4.11) = 125.19$ N

3 **[Foundational]** • **Target: Motion on Rough Inclined Planes**

Particles of mass 5 kg and 10 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 65.33. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 3.27$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 5(9.8 + 3.27) = 65.33$ N

4 **[Foundational]** • **Target: Connected Systems with Pulleys & Tables**

Particles of mass 2 kg and 8 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 31.36. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 5.88$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 2(9.8 + 5.88) = 31.36$ N

5 **[Foundational]** • **Target: Newton's Second Law ($F = ma$)**

Particles of mass 4 kg and 13 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 59.95. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 5.19$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 4(9.8 + 5.19) = 59.95$ N

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6 [Foundational] • Target: Connected Particles over a Pulley

Particles of mass 7 kg and 17 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 97.18. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 4.08$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 7(9.8 + 4.08) = 97.18$ N

7 [Foundational] • Target: Motion on Rough Inclined Planes

Particles of mass 7 kg and 16 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 95.44. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 3.83$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 7(9.8 + 3.83) = 95.44$ N

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Connected Systems with Pulleys & Tables

Particles of mass 6 kg and 7 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 63.32. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 0.75$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 6(9.8 + 0.75) = 63.32$ N

9 [Intermediate] • Target: Newton's Second Law ($F = ma$)

Particles of mass 4 kg and 11 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 57.49. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 4.57$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 4(9.8 + 4.57) = 57.49$ N

10 [Intermediate] • Target: Connected Particles over a Pulley

Particles of mass 2 kg and 5 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 28.00. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 4.20$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 2(9.8 + 4.20) = 28.00$ N

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11 [Intermediate] • Target: Motion on Rough Inclined Planes

Particles of mass 5 kg and 8 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 60.31. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 2.26$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 5(9.8 + 2.26) = 60.31$ N

12 [Intermediate] • Target: Connected Systems with Pulleys & Tables

Particles of mass 4 kg and 14 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 60.98. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 5.44$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = 4(9.8 + 5.44) = 60.98$ N

13 [Intermediate] • Target: Newton's Second Law ($F = ma$)

Particles of mass 1 kg and 7 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 17.15. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 7.35$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = (9.8 + 7.35) = 17.15$ N

14 [Intermediate] • Target: Connected Particles over a Pulley

Particles of mass 1 kg and 4 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Final Answer: 15.68. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = 5.88$ m/s^2. For mass m_1:

$T - m_1 g = m_1 a \Rightarrow T = m_1(g + a) = (9.8 + 5.88) = 15.68$ N

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Motion on Rough Inclined Planes • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Particles of mass 11 kg and 26 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

Final Answer: 3.97. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(26 - 11)(9.8)}{11 + 26} = 3.97$ m/s^2

16 [Challenging] • Target: Connected Systems with Pulleys & Tables • Cross-Lesson: 10.01 Series & Binomial Theorem

Particles of mass 12 kg and 14 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

Final Answer: 0.75. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(14 - 12)(9.8)}{12 + 14} = 0.75$ m/s^2

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17 [Challenging] • Target: Newton's Second Law ($F = ma$) • Cross-Lesson: 11.01 Kinematics

Particles of mass 6 kg and 11 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

Final Answer: 2.88. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(11-6)(9.8)}{6+11} = 2.88$ m/s^2

18 [Challenging] • Target: Connected Particles over a Pulley • Cross-Lesson: 12.01 Forces & Equilibrium

Particles of mass 8 kg and 10 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

Final Answer: 1.09. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(10-8)(9.8)}{8+10} = 1.09$ m/s^2

19 [Challenging] • Target: Motion on Rough Inclined Planes • Cross-Lesson: 14.01 Energy, Work & Power

Particles of mass 4 kg and 13 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

Final Answer: 5.19. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(13-4)(9.8)}{4+13} = 5.19$ m/s^2

20 [Challenging] • Target: Connected Systems with Pulleys & Tables • Cross-Lesson: 15.01 Representation of Data

Particles of mass 8 kg and 15 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

Final Answer: 2.98. $a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(15-8)(9.8)}{8+15} = 2.98$ m/s^2

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Newton's Second Law ($F = ma$)

A constant resultant force of 36 N acts on a mass of 4 kg. What is the acceleration produced?

a. 9 m/s^2 ✓

b. 144 m/s^2

c. 4 m/s^2

d. 6 m/s^2

$a = F/m = 36/4 = 9$ m/s^2.

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22 [Foundational] • Target: Newton's Second Law ($F = ma$)

A car of mass 1200 kg is traveling at 25 m/s when the brakes are applied, bringing it to rest in 50 m. Find the magnitude of the braking force.

a. 7500 N ✓

b. 6000 N

c. 9000 N

d. 15000 N

$v^2 = u^2 + 2as \Rightarrow 0 = 25^2 + 2a(50) \Rightarrow 100a = -625 \Rightarrow a = -6.25$ m/s². Force magnitude $F = ma = 1200(6.25) = 7500$ N.

23 [Foundational] • Target: Newton's Second Law ($F = ma$)

An elevator of mass 800 kg accelerates upwards at 1.5 m/s². Taking $g = 10$ m/s², find the tension in the cable.

a. 9200 N ✓

b. 8000 N

c. 6800 N

d. 1200 N

Equation of motion: $T - mg = ma \Rightarrow T = m(g + a) = 800(10 + 1.5) = 800(11.5) = 9200$ N.

24 [Foundational] • Target: Connected Particles over a Pulley

Two masses of 3 kg and 2 kg are connected by a light inextensible string passing over a smooth fixed pulley (Atwood machine). Taking $g = 10$ m/s², find the acceleration of the system.

a. 2 m/s² ✓

b. 1 m/s²

c. 4 m/s²

d. 5 m/s²

Equations: $3g - T = 3a$ and $T - 2g = 2a$. Adding: $(3 - 2)g = (3 + 2)a \Rightarrow 10 = 5a \Rightarrow a = 2$ m/s².

25 [Foundational] • Target: Connected Particles over a Pulley

For the system in the previous question, find the tension T in the string.

a. 24 N ✓

b. 25 N

c. 20 N

d. 30 N

$T = 2(g + a) = 2(10 + 2) = 2(12) = 24$ N.

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26 [Foundational] • Target: Connected Particles over a Pulley

What is the total force exerted by the string on the pulley in the previous system?

a. 48 N **downwards** ✓

b. 24 N downwards

c. 50 N downwards

d. 40 N downwards

The string pulls downward on both sides of the pulley with tension $T = 24$ N. Total force on pulley $= 2T = 48$ N downwards.

27 [Foundational] • Target: Motion on Rough Inclined Planes

A block of mass 5 kg is pulled up a smooth slope inclined at 30° to the horizontal by a force of 40 N acting parallel to the slope. Taking $g = 10$ m/s^2, find the acceleration of the block.

a. 3 m/s^2 ✓

b. 5 m/s^2

c. 2 m/s^2

d. 4 m/s^2

$F_{\text{net}} = P - mg\sin(30^\circ) = 40 - 5(10)(0.5) = 40 - 25 = 15$ N. Then $a = 15/5 = 3$ m/s^2.

28 [Foundational] • Target: Motion on Rough Inclined Planes

If the slope in the previous question is rough with $\mu = 0.2$, what is the new acceleration of the block?

a. 1.27 m/s^2 ✓

b. 3.00 m/s^2

c. 2.15 m/s^2

d. 0.85 m/s^2

$R = mg\cos(30^\circ) = 50(\sqrt{3}/2) \approx 43.30$ N. Friction $F_f = \mu R = 0.2(43.30) \approx 8.66$ N. Net force $= 40 - 25 - 8.66 = 6.34$ N. Acceleration $a = 6.34/5 \approx 1.27$ m/s^2.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Motion on Rough Inclined Planes

A sled of mass 20 kg slides down an incline of 25° at constant speed. What is the coefficient of kinetic friction μ?

a. $\mu = \tan(25^\circ) \approx 0.466$ ✓

b. 0.423

c. 0.906

d. 0.500

At constant speed, acceleration is zero, so $mg\sin\alpha = \mu mg\cos\alpha \Rightarrow \mu = \tan\alpha = \tan(25^\circ) \approx 0.466$.

30 [Intermediate] • Target: Connected Systems with Pulleys & Tables

A block A of mass 4 kg rests on a smooth horizontal table and is connected by a light string over a smooth pulley at the edge of the table to a hanging block B of mass 1 kg. Taking $g = 10$ m/s^2, find the acceleration.

a. 2 m/s^2 ✓

b. 2.5 m/s^2

c. 1.5 m/s^2

d. 10 m/s^2

Net accelerating force is the weight of B: $m_B g = 1(10) = 10$ N. Total mass $= 4 + 1 = 5$ kg. Acceleration $a = 10/5 = 2$ m/s^2.

31 [Intermediate] • Target: Connected Systems with Pulleys & Tables

If the table in the previous system has coefficient of friction $\mu = 0.15$, find the acceleration of the system.

a. 0.8 m/s^2 ✓

b. 1.2 m/s^2

c. 1.5 m/s^2

d. 0.5 m/s^2

Friction on A: $F_f = \mu m_A g = 0.15(4)(10) = 6$ N. Net accelerating force $= 10 - 6 = 4$ N. Total mass $= 5$ kg. Acceleration $a = 4/5 = 0.8$ m/s^2.

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32 [Intermediate] • Target: Connected Systems with Pulleys & Tables

A car of mass 1000 kg tows a trailer of mass 500 kg with acceleration 1.2 m/s^2. Resistance to motion is 300 N on the car and 200 N on the trailer. Find the tension in the towbar.

a. 800 N ✓

b. 600 N

c. 1100 N

d. 1500 N

Consider the trailer alone: $T - R_{\text{trailer}} = m_{\text{trailer}}a \Rightarrow T - 200 = 500(1.2) \Rightarrow T - 200 = 600 \Rightarrow T = 800$ N.

33 [Intermediate] • Target: Newton's Second Law ($F = ma$)

Particles of mass 5 kg and 17 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 5.85

b. **5.35** ✓

c. 10.69

d. 4.85

$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(17 - 5)(9.8)}{5 + 17} = 5.35$ m/s^2

34 [Intermediate] • Target: Connected Particles over a Pulley

Particles of mass 9 kg and 15 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 4.90

b. **2.45** ✓

c. 1.95

d. 2.95

$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(15 - 9)(9.8)}{9 + 15} = 2.45$ m/s^2

35 [Intermediate] • Target: Motion on Rough Inclined Planes

Particles of mass 2 kg and 5 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. **4.20** ✓

b. 8.40

c. 4.70

d. 3.70

$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(5 - 2)(9.8)}{2 + 5} = 4.20$ m/s^2

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36 [Intermediate] • Target: Connected Systems with Pulleys & Tables

Particles of mass 1 kg and 3 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. **4.90** ✓

b. 9.80

c. 5.40

d. 4.40

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(3-1)(9.8)}{1+3} = 4.90 \text{ m/s}^2$$

37 [Intermediate] • Target: Newton's Second Law ($F = ma$)

Particles of mass 3 kg and 4 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 0.90

b. **1.40** ✓

c. 1.90

d. 2.80

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(4-3)(9.8)}{3+4} = 1.40 \text{ m/s}^2$$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Connected Particles over a Pulley • Cross-Lesson: 12.01 Forces & Equilibrium

Particles of mass 1 kg and 4 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 5.38

b. 11.76

c. **5.88** ✓

d. 6.38

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(4-1)(9.8)}{1+4} = 5.88 \text{ m/s}^2$$

39 [Challenging] • Target: Motion on Rough Inclined Planes • Cross-Lesson: 14.01 Energy, Work & Power

Particles of mass 5 kg and 9 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 3.30

b. 2.30

c. 5.60

d. **2.80** ✓

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(9-5)(9.8)}{5+9} = 2.80 \text{ m/s}^2$$

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40 [Challenging] • Target: Connected Systems with Pulleys & Tables • Cross-Lesson: 15.01 Representation of Data

Particles of mass 2 kg and 8 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 5.38

b. **5.88** ✓

c. 6.38

d. 11.76

$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(8-2)(9.8)}{2+8} = 5.88$ m/s^2

41 [Challenging] • Target: Newton's Second Law ($F = ma$) • Cross-Lesson: 16.01 Permutations & Combinations

Particles of mass 4 kg and 8 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 3.77

b. 6.53

c. 2.77

d. **3.27** ✓

$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(8-4)(9.8)}{4+8} = 3.27$ m/s^2

42 [Challenging] • Target: Connected Particles over a Pulley • Cross-Lesson: 17.01 Discrete Random Variables

Particles of mass 4 kg and 9 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 3.27

b. **3.77** ✓

c. 4.27

d. 7.54

$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(9-4)(9.8)}{4+9} = 3.77$ m/s^2

43 [Challenging] • Target: Motion on Rough Inclined Planes • Cross-Lesson: 18.01 Binomial Distribution

A particle of mass 14 kg slides down a rough plane inclined at 60° to the horizontal, experiencing a constant friction force of 7 N. Taking $g = 9.8$ m/s^2, find its acceleration down the slope.

a. 15.97

b. 8.49

c. **7.99** ✓

d. 7.49

$W = mg = 14 \times 9.8 = 137.20$ N. Component along slope $= W\sin 60^\circ \approx 118.82$ N. Net force ≈ 111.82 N. $a = F/m \approx 7.99$ m/s^2

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44 [Challenging] • Target: Connected Systems with Pulleys & Tables • Cross-Lesson: 19.01 Normal Distribution

A particle of mass 12 kg slides down a rough plane inclined at 30° to the horizontal, experiencing a constant friction force of 9 N. Taking $g = 9.8$ m/s^2, find its acceleration down the slope.

a. 3.65

b. **4.15** ✓

c. 8.30

d. 4.65

$W = mg = 12 \times 9.8 = 117.60$ N. Component along slope $= W\sin 30^\circ \approx 58.80$ N. Net force ≈ 49.80 N. $a = F/m \approx 4.15$ m/s^2

45 [Challenging] • Target: Newton's Second Law ($F = ma$) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A particle of mass 12 kg slides down a rough plane inclined at 30° to the horizontal, experiencing a constant friction force of 3 N. Taking $g = 9.8$ m/s^2, find its acceleration down the slope.

a. 4.15

b. **4.65** ✓

c. 5.15

d. 9.30

$W = mg = 12 \times 9.8 = 117.60$ N. Component along slope $= W\sin 30^\circ \approx 58.80$ N. Net force ≈ 55.80 N. $a = F/m \approx 4.65$ m/s^2

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